

Sachin Ranjan

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GitHub — LinkedIn — LeetCode

Professional Summary

Computer Science undergraduate pursuing a B.Tech in Computer Science & Engineering with strong foundations in Java, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, and Software Engineering. Experienced in developing AI-powered applications using Machine Learning and Computer Vision. Passionate about Backend Development, REST APIs, scalable software systems, and solving real-world engineering problems through clean, efficient, and production-ready code. Actively preparing for Software Development Engineer (SDE) roles at leading product-based companies.

Education

Quantum University, Roorkee

2024 – 2028

Bachelor of Technology (Computer Science & Engineering)

CGPA: 8.20/10

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming

Technical Skills

- **Programming Languages:** Java, Python, C++, C, SQL
- **Backend Development:** JDBC, Maven, Spring Boot (Learning), REST APIs (Learning)
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **Databases:** MySQL, PostgreSQL
- **AI / Machine Learning:** Machine Learning, Deep Learning, Computer Vision, YOLOv12, CNN, Random Forest
- **Developer Tools:** Git, GitHub, Linux, IntelliJ IDEA, VS Code, Streamlit

Projects

Mushroom Detection & Classification System

Tech Stack: Python, YOLOv12, OpenCV, Streamlit, Deep Learning

- Developed an AI-powered mushroom detection and classification system using YOLOv12.
- Built a complete image preprocessing and real-time inference pipeline.
- Designed an interactive Streamlit web application for image upload and prediction.
- Improved prediction performance through deep learning optimization techniques.

AI Crop Recommendation System

Tech Stack: Python, Random Forest, Scikit-learn, Pandas, NumPy, Streamlit

- Developed a Machine Learning-based crop recommendation system using Random Forest.

- Predicted suitable crops based on soil nutrients and environmental conditions.
- Built a user-friendly Streamlit application for real-time recommendations.
- Improved agricultural decision-making through intelligent prediction.

Publication

- Published the research paper titled ”**Detection and Classification of Edible Mushrooms Using Deep Learning (YOLOv12)**” at IEEE UPWIECON 2025.

Achievements

- Published research paper at IEEE UPWIECON 2025.
- Earned NPTEL Elite + Silver Certification in Programming in Java.
- Achieved 5-Star C++ on HackerRank.

Certifications

- Earned **NPTEL Elite + Silver Certification** in Programming in Java from **IIT Kharagpur**.
- Quantum Computing — IIT Roorkee
- C++ Programming

Coding Profiles

GitHub

LeetCode

HackerRank

LinkedIn